Two Topics in Wave Optics

Phasor diagrams and Fourier optics

Nikola Kočović & Luka Smiljanić September 2026

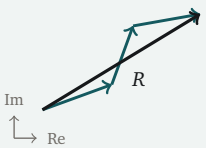
High school wave optics usually means a handful of fixed setups: a single slit, a double slit, a grating. Olympiad problems throw far stranger apertures at you, and the brute-force method — integrating the wavelets radiated from every point of the aperture — gets messy. This handout collects two ways around that integral: *phasors* (usually used in AC circuits), for reading the intensity distribution of a few basic apertures straight off a diagram; and *Fourier optics*, for building any complicated aperture's pattern from simple pieces with the convolution theorem, and for getting an obstacle's pattern from its complement's with Babinet's principle.

It states results and moves fast; every derivation it skips is worked out in full in Appendix A.

1 PHASORS: READING OFF INTENSITY

I presume you are already familiar with the concept of phasors and phasor diagrams; if not, use a source like Kalda's *Wave Optics* or Kevin Zhou's *Waves* handout.

idea 1 *the phasor picture* Chain the phasors head to tail; the length and angle of the arrow from start to finish give the amplitude and phase of the total field — square that length for the intensity, $I \propto R^2$.

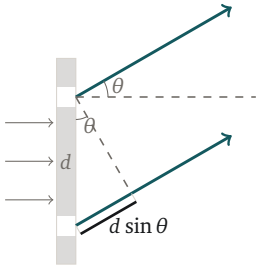


$k_x = k \sin\theta$ is the transverse wavevector for a ray leaving the aperture at angle θ — the same k_x that reappears as the Fourier transform's frequency in Section 2.1. We use the notation k_x here in the phasor part too, precisely to make it visually obvious that both methods describe the same physics.

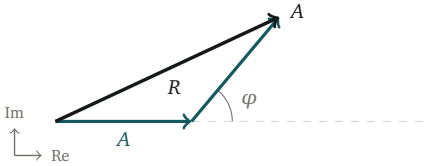
1.1 *The double slit: two phasors*

EXAMPLE *the double slit* Two thin slits, separated by d, are lit normally by light of wavelength λ. Find the resultant amplitude and intensity as a function of k_x.

SOLUTION In the far field the two outgoing rays are essentially parallel, tilted by θ from the normal, and the ray from the far slit has an extra distance $d \sin\theta$ to travel.



That extra path is a phase $\varphi = k(d\sin\theta) = k_x d$ behind the near slit — so as phasors, the two contributions have equal length A, pointing φ apart.



Using simple trigonometry, we get

$$R = 2A\cos\left(\frac{\varphi}{2}\right),$$

and since $\varphi = k_x d$,

fact 1 *double slit* Two thin slits a distance d apart give a resultant

$$R(k_x) = 2A\cos\left(\frac{k_x d}{2}\right),$$

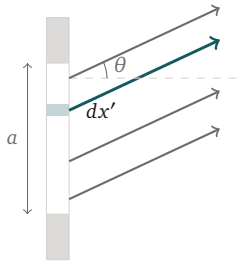
so the intensity is $I(k_x) \propto R(k_x)^2 = 4A^2\cos^2(k_x d/2)$.

NOTE That's the same result we'll get via integration in fact 4.

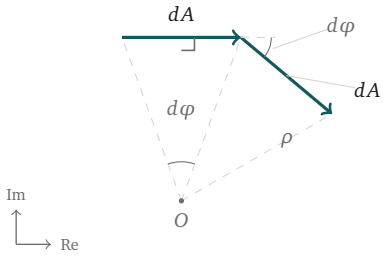
1.2 *The wide slit: an arc of phasors*

EXAMPLE *the wide slit* A single slit of width a is lit normally by light of wavelength λ. Find the resultant amplitude and intensity as a function of k_x.

SOLUTION Let's say we have one slit with width a.



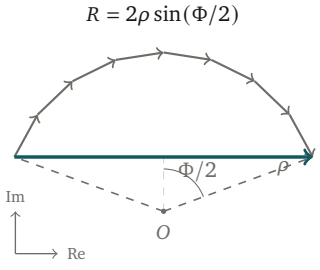
Split it into infinitesimal strips dx': strip x' radiates a phasor of length dA (proportional to dx'), turned by a tiny extra phase $d\varphi = k_x\, dx'$ from its neighbour:



Therefore, two neighbouring phasors meet at the small angle $d\varphi$. Together they form a triangle — call O the center of its circumscribed circle, and ρ that circle's radius, the arc's radius of curvature. O and either phasor form a thin isosceles triangle of apex angle $d\varphi$; bisecting it gives two right triangles with opposite side $dA/2$ and adjacent side $\approx \rho$, so

$$\frac{dA}{2} = \rho \tan\!\left(\frac{d\varphi}{2}\right) \approx \rho\,\frac{d\varphi}{2} \qquad (d\varphi \to 0),$$

i.e. $\rho = dA/d\varphi$ — the same all along the slit, since both dA and $d\varphi$ scale with dx'. A constant radius of curvature is exactly a circular arc, of total length $L = \int dA$ and total angle $\Phi = \int d\varphi = k_x a$. Then with some more basic trigonometry (see figure):



$\operatorname{sinc}(u) = \sin u/u.$

$$R = 2\rho \sin\!\left(\frac{\Phi}{2}\right) = L\,\operatorname{sinc}\!\left(\frac{\Phi}{2}\right),$$

and since $\Phi = k_x a$ and $L = A$,

fact 2 *wide slit* A slit of width a gives a resultant

$$R(k_x) = A\ \mathrm{sinc}\left(\frac{k_x a}{2}\right),$$

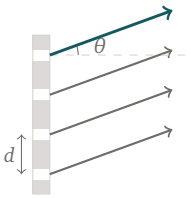
so the intensity is $I(k_x) \propto R(k_x)^2 = A^2\,\mathrm{sinc}^2(k_x a/2)$.

Dots give difficulty; points, time and a source are optional.

1 •○○○ N slits, by phasors

2 pts · 15 min

N identical narrow slits, spaced by d, are lit normally by light of wavelength λ.



N slits, spaced by d

Using a phasor diagram, show that

$$I(k_x) \propto \left[\frac{\sin(Nk_x d/2)}{\sin(k_x d/2)}\right]^2.$$

NOTE This works cleanly for setups this simple, but for anything more complicated it gets too clunky — that is exactly why we need the second part of this handout.

2 FOURIER OPTICS

2.1 *Diffraction as a Fourier transform*

def. 1 *Fraunhofer diffraction* For an aperture with transmission function $t(x)$, the diffracted complex amplitude at transverse spatial frequency k_x is

$$U(k_x) = \int_{-\infty}^{\infty} t(x)\, e^{-ik_x x}\, \mathrm{d}x,$$

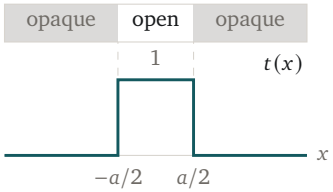
and the measured intensity is $I(k_x) \propto |U(k_x)|^2$.

The integral sums up the waves from every point of the aperture. This makes $U(k_x)$ proportional to the Fourier transform of the transmission

function, $U(k_x) \propto \mathcal{F}\{t(x)\}(k_x)$ — the missing prefactor is a constant, so it never changes the shape of the pattern.

Physically, $t(x)$ is just a rule attached to each point of the aperture, saying what happens to the wave there: $t = 0$ blocks it, $t = 1$ passes it unchanged, a number in between is a grey partly-absorbing filter, and a complex $t = e^{i\phi}$ of modulus 1 passes it fully but delays it by a phase ϕ (a sheet of glass, say) — so in general $t(x) = A(x)e^{i\phi(x)}$, amplitude and phase set independently at every point.

The simplest case is a single gap of width a cut in an opaque screen: unchanged inside the gap, blocked outside it, i.e. $t(x) = \mathrm{rect}(x/a)$.



A single slit of width a has $t(x) = \mathrm{rect}(x/a)$, and a direct integration gives $U(k_x) = a\,\mathrm{sinc}(k_x a/2)$ (Appendix A.2).

$\mathrm{rect}\left(\frac{x-x_0}{a}\right) = 1$ for $|x - x_0| < a/2$ and 0 otherwise.

2.2 *Convolution*

def. 2 *convolution* For two functions f, g,

$$(f * g)(x) = \int_{-\infty}^{\infty} f(\tau)\, g(x - \tau)\, \mathrm{d}\tau.$$

Full derivation in Appendix A.1.

fact 3 *convolution theorem*

$$\mathcal{F}\{f * g\} = \mathcal{F}\{f\} \cdot \mathcal{F}\{g\}.$$

idea 2 *build, don't integrate* Write a complicated aperture as a convolution of simple pieces. By fact 3 its Fourier transform is just the product of their transforms, so a hard integral becomes several easy ones.

fact 4 *Dirac delta* $\int f(x)\delta(x - x_0)\,\mathrm{d}x = f(x_0)$, and consequently $f(x) * \delta(x - x_0) = f(x - x_0)$: convolving with a shifted delta

Also $\mathcal{F}\{\delta(x - x_0)\} = e^{-ik_x x_0}$: a shift in real space is a phase factor in Fourier space (Appendix A.3).

translates a pattern without changing its shape.

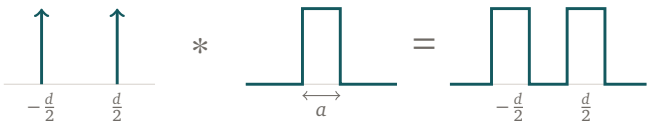
2.3 *Worked example: the double slit, assembled*

EXAMPLE *double slit as a convolution* Two slits of width a, separated by d. Write

$$t(x) = \left[\delta\left(x - \tfrac{d}{2}\right) + \delta\left(x + \tfrac{d}{2}\right)\right] * \mathrm{rect}(x/a)$$

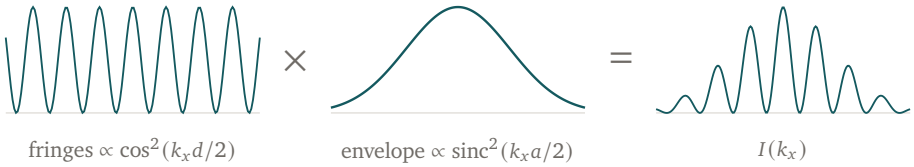
and find $I(k_x)$.

SOLUTION We use delta functions to position things. Convolving with a sum of shifted deltas just stacks a copy of the other function at each delta's position (fact 4):



The two-delta factor transforms to $F_1(k_x) = 2\cos(k_x d/2)$ (fact 4), the rect factor to $F_2(k_x) = a\,\mathrm{sinc}(k_x a/2)$ (Section 2.1), so $U = F_1 F_2$ gives

$$I(k_x) \propto 4a^2 \cos^2\!\left(\frac{k_x d}{2}\right) \mathrm{sinc}^2\!\left(\frac{k_x a}{2}\right)$$

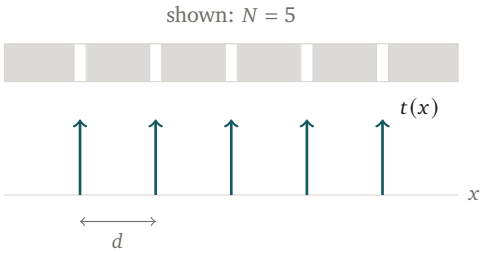


NOTE The $\cos^2$ fringes (from the separation d) and the sinc^2 envelope (from the width a) each came from a one-line transform — exactly idea 2.

2 ●●○○ N narrow slits, by Fourier

3 pts

N identical, infinitely narrow slits, cut in an opaque screen and spaced by d, are lit normally by light of wavelength λ, so $t(x) = \sum_{n=0}^{N-1} \delta(x - nd)$. You solved this once already with a phasor diagram (Section 1) — now do it with Fourier transforms and check the two answers agree.[h2]



Show that

$$I(k_x) \propto \left[\frac{\sin(Nk_x d/2)}{\sin(k_x d/2)}\right]^2,$$

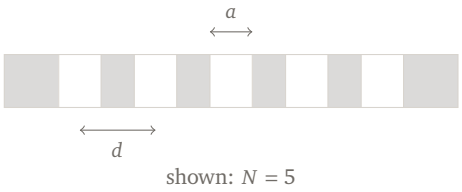
and describe how the width and sharpness of the principal maxima depend on N.

3 ●●○○ N slits, finite width

2 pts · 15 min

Same N slits, spaced by d, as above — but now give each one a finite width $a < d$, so[h3]

$$t(x) = \mathrm{rect}(x/a) * \sum_{n=0}^{N-1} \delta(x - nd).$$



Show that

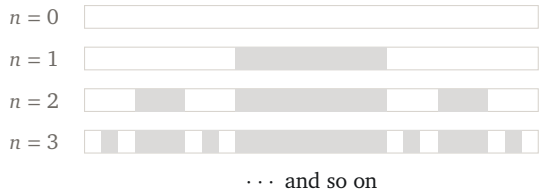
$$I(k_x) \propto \mathrm{sinc}^2\!\left(\frac{k_x a}{2}\right)\left[\frac{\sin(Nk_x d/2)}{\sin(k_x d/2)}\right]^2,$$

and say which factor sets the spacing of the bright fringes and which sets their overall envelope.

4 ●●●○ Cantor interference

3 pts · 30 min

A slit of width $d = 1\,\mathrm{cm}$ is divided into three equal parts[h4.1] with the middle third covered; the same is done to each of the two remaining slits, and so on for many iterations. Light of wavelength $\lambda = 500\,\mathrm{nm}$ illuminates the resulting aperture, producing fringes on a screen $L = 10\,\mathrm{m}$ away. Find the distance x between the central maximum and the first side maximum, in millimetres.[h4.2]



2.4 *A shift is a phase*

A time delay Δt on a monochromatic wave of frequency ω is a phase shift $\Delta\phi = -\omega\Delta t$.

idea 3 *delaying one slit* Replace one delta above by $e^{i\Delta\phi}\delta(x+d/2)$ (fact 4 with a phase attached) to model a delayed slit. Then $U(k_x) = e^{-ik_x d/2} + e^{i\Delta\phi}e^{ik_x d/2}$, so

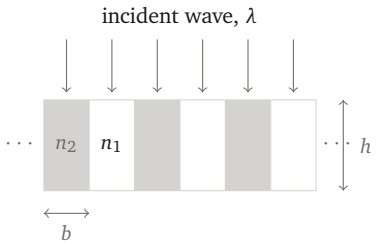
$$I(k_x) \propto 2 + 2\cos(k_x d + \Delta\phi) :$$

a delay shifts the fringe pattern in phase, not in spacing. This is the delta-function case of the general shift theorem, Appendix A.4.

5 ●●●○ Binary phase diffraction grating

3 pts · 45 min · *Bulgaria TST 2022*

A slab of thickness h is made of alternating strips of width b and refractive indices n_1, n_2, total width $2Nb$ with $N \gg 1$. A parallel monochromatic beam of wavelength λ hits it normally; neglect reflections at the surfaces.[h5.1]



(a) Find the minimum thickness $h_{\min}$ for which $I(\Theta = 0) = 0$.[h5.2] Use $h = h_{\min}$ below.

(b) Find the angle Θ_1 of the first diffraction maximum.

(c) Find the angle Θ_2 of the second diffraction maximum.

(d) Find the general intensity pattern $I(\Theta)$.

NOTE The next problem is two-dimensional. For an aperture $t(x,y)$ the far field is $U(k_x,k_y) = \iint t(x,y)\,e^{-i(k_x x + k_y y)}\,dx\,dy$. If the aperture is separable, $t(x,y) = t_x(x)\,t_y(y)$, the double integral factorises into $U = U_x(k_x)\,U_y(k_y)$, so the pattern is the product of two one-dimensional patterns.

6 •••• Checkerboard reflective grating

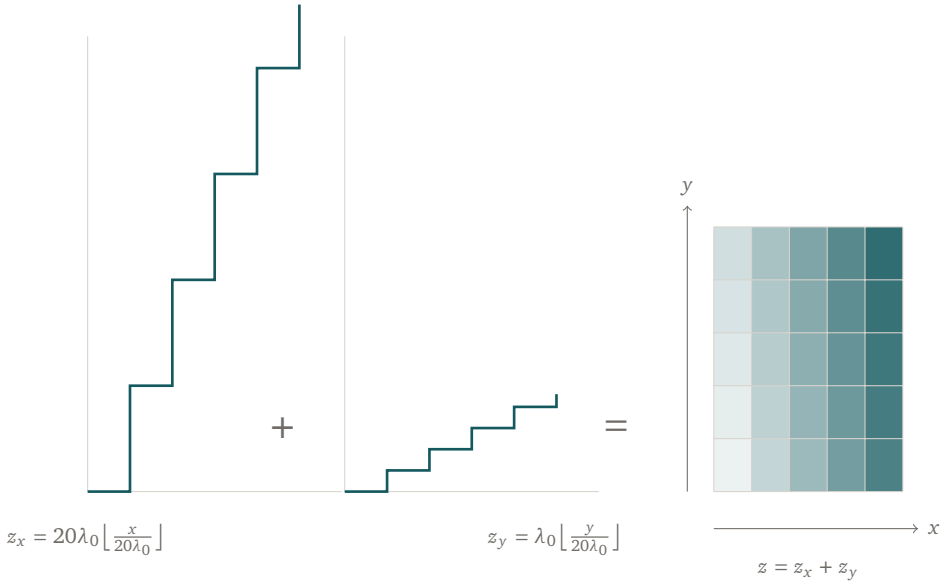
5 pts · 45 min

A reflecting surface has height

$$z = 20\lambda_0 \left\lfloor \frac{x}{20\lambda_0} \right\rfloor + \lambda_0 \left\lfloor \frac{y}{20\lambda_0} \right\rfloor, \qquad \lambda_0 = 500\,\text{nm},$$

over $|x|, |y| < 5000\lambda_0$ (a 5 mm × 5 mm square); the vertical step faces are black. White light ($400\,\text{nm} \le \lambda \le 700\,\text{nm}$) falls on it along z, and a lens focuses the reflected light onto a screen 50 cm away.[h6.1]

The height is a sum of two independent staircases, one in x and one in y, both with the same $20\lambda_0$ tread width but very different riser heights:[h6.2]



Darker = taller terrace. Each cell is $20\lambda_0 \times 20\lambda_0$; the grid continues far beyond what's drawn, since the true grating is 5 mm × 5 mm across.

(a) Sketch the pattern on the screen.

(b) Estimate its dimensions.

(c) For each wavelength, find how the position of the brightest spot depends on λ.

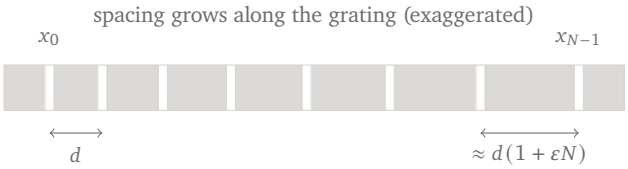
7 •••• A grating with a worn lead screw

5 pts · 60 min

A ruling engine cuts N identical, infinitely narrow slits at positions

$$x_n = nd\left(1 + \tfrac{\varepsilon}{2}\, n\right), \qquad n = 0, \ldots, N-1, \quad \varepsilon N \ll 1,$$

so the period grows linearly along the grating. It is lit normally by light of wavelength λ, and the far field is observed near the m-th order, $k_x = 2\pi m/d + \kappa$ with $|\kappa| d \ll 2\pi$.



(a) Show that, to leading order in εN,[h7.1]

$$U(\kappa) \simeq \sum_{n=0}^{N-1} \exp\!\big[-\mathrm{i}\big(\pi m \varepsilon\, n^2 + \kappa d\, n\big)\big].$$

(b) Define $N_c = 1/\sqrt{m\varepsilon}$. For $N \ll N_c$ find the peak intensity. For $N \gg N_c$ find the shape of the order, its peak intensity, and its width.[h7.2] Explain it physically in terms of the local period. What happens to the $m = 0$ order?

(c) Show that the integrated power in the order is independent of ε. Find the ratio of the intensity at the edge of the plateau to its height.

(d) A spectrometer uses this grating to resolve two close lines. Estimate its resolving power as a function of N and show that there is an optimal N beyond which ruling more slits makes the instrument worse.[h7.3]

2.5 *Babinet's principle*

Some apertures are hard to write down directly — a thin wire, a hair, an irregular obstacle — but easy to write down as the *complement* of an aperture we've already solved.

def. 3 *complementary apertures* Two transmission functions are complementary if $t_1(x) + t_2(x) = 1$: wherever one is open the other is blocked.

idea 4 *Babinet's principle* A constant $t = 1$ (no aperture at all) has Fourier transform zero everywhere except at $k_x = 0$ (Appendix A.3 with $x_0 = 0$). So for $k_x \neq 0$, def. 3 and linearity give $U_1(k_x) + U_2(k_x) = 0$: complementary apertures give the same diffraction pattern, except in the forward direction.

fact 5 *Babinet's principle* If $t_1 + t_2 = 1$, then $I_1(k_x) = I_2(k_x)$ for every $k_x \neq 0$.

EXAMPLE *diffraction by a thin wire* A thin opaque wire of width a, much narrower than it is long, blocks part of a beam of light of wavelength λ. Find the intensity pattern observed away from the forward direction.

SOLUTION The wire's complement is open everywhere except a strip of width a — that's the slit from fact 2. By fact 5 the two patterns agree for $k_x \neq 0$:

$$I(k_x) \propto \operatorname{sinc}^2\!\left(\frac{k_x a}{2}\right), \qquad k_x \neq 0.$$

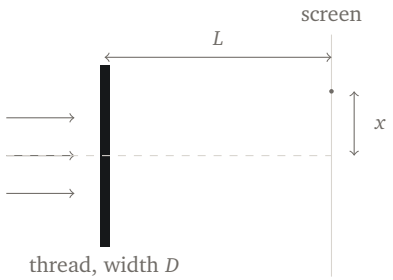
So the wire gives the same fringe pattern as a slit of the same width; fact 2's derivation never used whether the strip was open or blocked.

This is the classic way to measure a hair's or fibre's diameter without a microscope.

8 •••• diameter of a thread

3 pts · 15 min

A thin thread hangs in a laser beam ($\lambda = 650\,\text{nm}$), much wider than the thread, and casts a diffraction pattern on a screen $L = 2\,\text{m}$ behind it. The first dark fringe, on either side of the central bright band, falls a distance $x = 1.3\,\text{cm}$ from the center.



(a) Find the thread's diameter D, and check that the small-angle approximation you used is justified.

(b) The thread is replaced by a slit of the same width D in an opaque screen. Where does the pattern on the screen differ from the thread's, and why?

9 •••• Hairs on a window

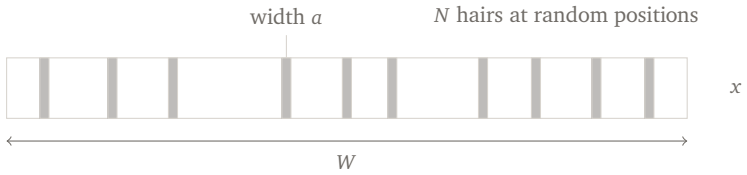
4 pts · 45 min

A plane wave of unit amplitude illuminates a window of width W. Stuck to it are $N \gg 1$ identical opaque strips ("hairs") of width a, with positions x_n independent and uniformly distributed over the window. Let $c = Na/W \ll 1$ be the fraction of the window they cover, and neglect overlaps,

so

$$t(x) = 1 - \sum_n \operatorname{rect}\!\left(\frac{x - x_n}{a}\right).$$

Take $W \gg a \gg \lambda$, work in the far field, and let $\langle\cdot\rangle$ denote the average over the positions x_n.



(a) Find the averaged intensity $\langle I(k_x)\rangle$, including the region near $k_x = 0$. Split it into a coherent part $|\langle U\rangle|^2$ and an incoherent "halo" $\langle |U|^2\rangle - |\langle U\rangle|^2$, and find both.[h9.1] Which of them survives for $|k_x| \gg 1/W$?

(b) Integrate each part over k_x to get the power in the forward beam and in the halo, as fractions of the incident power, to first order in c.[h9.2]

(c) The hairs block a fraction c of the geometric cross-section. Show that the forward beam loses twice that, and explain where the extra power goes.[h9.3]

NOTE Three techniques did all the work here: phasors read off intensity directly for simple sums of sources, the convolution theorem builds a complicated aperture's transform from the transforms of simple pieces, and Babinet's principle reuses that same result to get an obstacle's pattern from its complement's. Together they turn most diffraction integrals into algebra — go back to def. 1 and integrate directly only when none of them apply.

A DERIVATIONS

A.1 *Proof of the convolution theorem*

Starting from def. 2, take the Fourier transform of $f * g$ and swap the order of integration:

$$\mathcal{F}\{f*g\}(k) = \int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} f(\tau)\, g(x-\tau)\,\mathrm{d}\tau\right] e^{-ikx}\,\mathrm{d}x = \int_{-\infty}^{\infty} f(\tau)\left[\int_{-\infty}^{\infty} g(x-\tau)\, e^{-ikx}\,\mathrm{d}x\right]\mathrm{d}\tau.$$

Substituting $u = x - \tau$ in the inner integral,

$$\int_{-\infty}^{\infty} g(x-\tau)\, e^{-ikx}\,\mathrm{d}x = e^{-ik\tau}\int_{-\infty}^{\infty} g(u)\, e^{-iku}\,\mathrm{d}u = e^{-ik\tau}\,\mathcal{F}\{g\}(k),$$

so

$$\mathcal{F}\{f*g\}(k) = \mathcal{F}\{g\}(k)\int_{-\infty}^{\infty} f(\tau)\, e^{-ik\tau}\,\mathrm{d}\tau = \mathcal{F}\{f\}(k)\,\mathcal{F}\{g\}(k),$$

which is fact 3.

A.2 *Fourier transform of a rectangle*

For $t(x) = \mathrm{rect}(x/a)$,

$$U(k_x) = \int_{-a/2}^{a/2} e^{-ik_x x}\,\mathrm{d}x = \frac{e^{-ik_x a/2} - e^{ik_x a/2}}{-ik_x} = \frac{2\sin(k_x a/2)}{k_x} = a\ \mathrm{sinc}\left(\frac{k_x a}{2}\right),$$

using $e^{-i\theta} - e^{i\theta} = -2i\sin\theta$.

A.3 *The Dirac delta*

$\delta(x)$ vanishes for $x \neq 0$ and is normalized by $\int_{-\infty}^{\infty} \delta(x)\,\mathrm{d}x = 1$; together these give the sifting property $\int f(x)\delta(x - x_0)\,\mathrm{d}x = f(x_0)$ used in fact 4. Its transform follows directly:

$$\mathcal{F}\{\delta(x - x_0)\} = \int_{-\infty}^{\infty} \delta(x - x_0)\, e^{-ik_x x}\,\mathrm{d}x = e^{-ik_x x_0}.$$

A.4 *The shift theorem*

More generally, for any f, substituting $u = x - x_0$ gives

$$\mathcal{F}\{f(x{-}x_0)\}(k) = \int_{-\infty}^{\infty} f(x{-}x_0)\, e^{-ikx}\,\mathrm{d}x = e^{-ikx_0} \int_{-\infty}^{\infty} f(u)\, e^{-iku}\,\mathrm{d}u = e^{-ikx_0}\,\mathcal{F}\{f\}(k).$$

Section 2.4's phase-delay result is the special case $f = \delta$.

HINTS

2. $U(k_x)$ is a finite geometric series in $e^{-ik_x d}$.

3. Convolve the previous problem's array with a single slit of width a.

4.1. Each iteration's aperture is the previous one convolved with two suitably spaced, scaled deltas.

4.2. You'll get a rapidly converging series — keep only its first two terms.

5.1. The grating is periodic with period $2b$: write one period as two adjacent phase-shifted rects, then convolve with a Dirac comb of spacing $2b$.

5.2. The two materials' contributions must cancel exactly — find the phase difference between them that does this.

6.1. Reflection doubles the path difference: the relevant phase is $2kz$, not kz.

6.2. Since $z = z_x + z_y$, the phase factor separates, $e^{2ikz} = e^{2ikz_x(x)}\, e^{2ikz_y(y)}$, so the transmission is separable. Find the pattern of each 1D staircase and multiply.

7.1. Expand $k_x x_n$: one term is a multiple of 2π for every n, and another is smaller than the rest by a factor κd.

7.2. For $N \gg N_c$ replace the sum by an integral over n. The phase is stationary at one value of n for each κ, so each κ receives light from one region of the grating.

7.3. The resolving power is the position of the order divided by its width. Use the width from (b) in both regimes.

9.1. Write U as the open-window term minus $a\,\mathrm{sinc}(k_x a/2)\sum_n e^{-ik_x x_n}$. For independent x_n, the average of the sum is N times one average, and $\langle|\sum_n e^{-ik_x x_n}|^2\rangle$ has N diagonal terms and $N(N-1)$ cross terms.

9.2. Normalise with Parseval: for the open window, $\int |U|^2\,\mathrm{d}k_x/2\pi = W$.

9.3. The hairs absorb exactly the fraction c. Check that forward beam, halo and absorbed power add up to the incident power.